Technical Document #851: Retrieval Augmented Generation - Comprehensive Overvi

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Query decomposition breaks complex queries into simpler sub-queries. It improves retrieval quality for multi-faceted questions.

Vector databases store dense embeddings for semantic search. Pinecone, Weaviate, and ChromaDB are popular vector database solutions.

Chunking splits documents into smaller segments for embedding. Optimal chunk size balances context retention with retrieval precision.

Retrieval-Augmented Generation (RAG) combines information retrieval with text generation. It grounds LLM responses in factual, retrieved documents.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Cybersecurity threats continue to evolve in complexity and scale. Organizations must invest in robust security measures to protect sensitive data.

Digital twins are virtual replicas of physical systems. They enable simulation and optimization without risking real-world resources.

Supply chain management leverages technology to optimize logistics and distribution. Real-time tracking and predictive analytics improve efficiency.

The rapid advancement of technology has transformed how organizations approach problem-solving and innovation. Companies must adapt to stay competitive in an ever-evolving landscape.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Key Takeaways:

- Vector databases store dense embeddings for semantic search.
- Hallucination detection identifies when LLMs generate factually incorrect information.
- Retrieval-Augmented Generation (RAG) combines information retrieval with text generation.